

5G NR Multicell Twin Implementation

One artefact, built in four stages

1 **Build the San Francisco scene**

Vexcel 3D mesh of a 1 km² downtown tile, three towers placed on it, and a street-level receiver grid.

2 **Compute propagation three ways**

A 3GPP formula map, a SionnaRT ray-traced map, and the ray-traced map with vegetation loss added.

3 **Add mobility and handoffs**

Six moving users are scored against every sector, so serving cell, handoff region, Doppler and ICI update together.

4 **Embed everything in one file**

The precomputed grids are embedded in a single HTML page. Changing any control re-renders automatically – no server, no re-run of the ray tracer.

The multicell layout is three towers over one real tile

- **San Francisco, downtown** – centre 37.801105, -122.404175 (latitude, longitude), 1 km² tile, the same tile as the earlier single-tower twin.
- **Three towers** A, B and C, 30 m above local ground, 500–559 m apart, each clearing its local roof by 18–29 m.
- **Three sectors per tower** at 90°, 210° and 330°, 4×4 array, EIRP 47 dBm at boresight, tilts computed 0–10°.
- **Receivers:** 25×25 grid at 40 m spacing, street level at ground + 1.5 m; points inside a building are nudged to the nearest open cell.



Three modes, one map, so the models can be compared fairly

3GPP

The model map. Simplified TR 38.901 LOS path loss – UMa $28.0 + 22 \cdot \log_{10}(d) + 20 \cdot \log_{10}(f)$ – computed live in the page at the selected frequency and morphology. No building shadowing, so it reads as a planning upper bound.

SionnaRT

The ray-traced map. Sionna 2.2.0 traces the Vexcel mesh in ITU concrete: LOS, specular, diffuse and diffraction paths, max depth 3, 1,000,000 samples per source. Grids are precomputed offline, then embedded – RSRP = EIRP - path loss, the same convention as 3GPP mode.

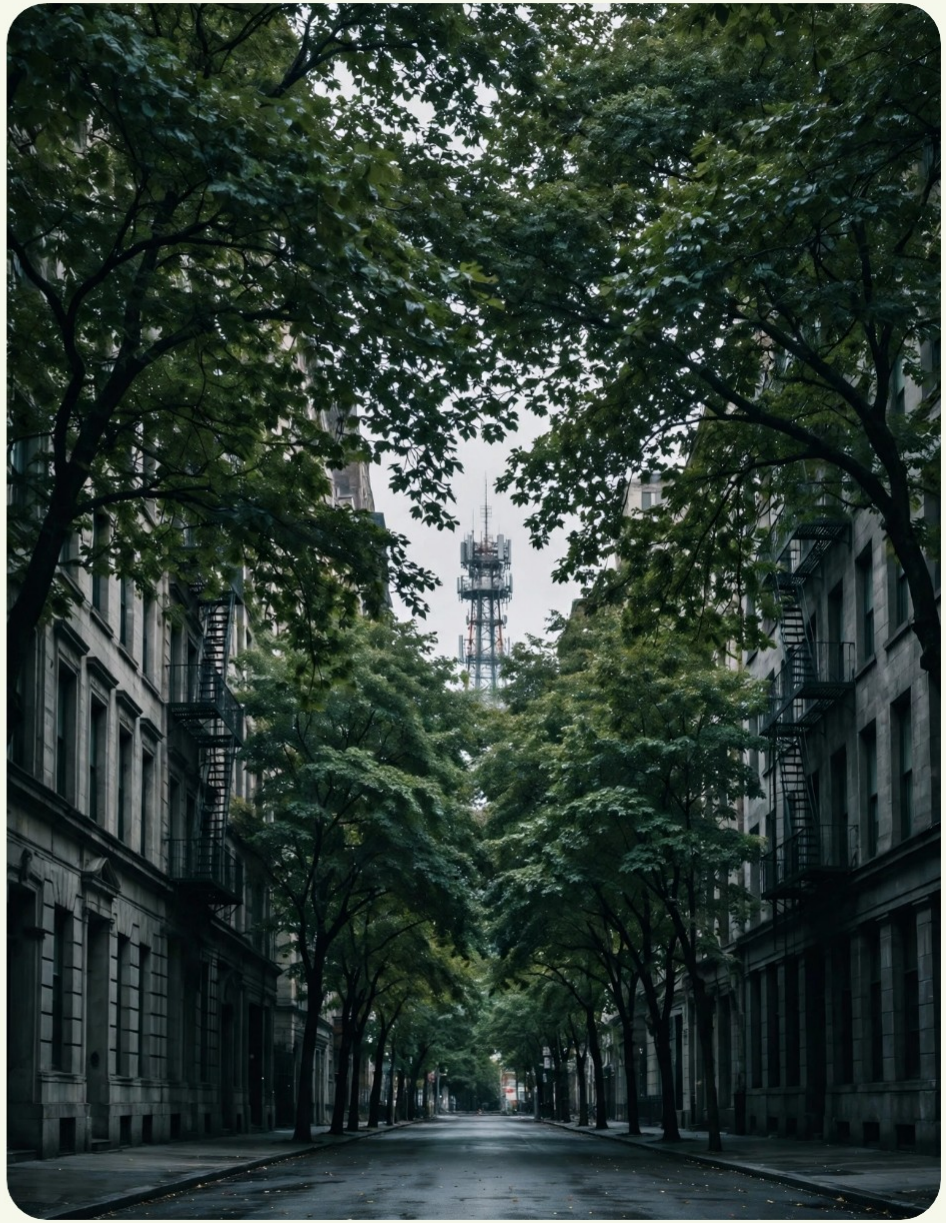
SionnaRT-Veg

The ray-traced map plus trees. The same SionnaRT grid with P.833 foliage loss subtracted per tower–receiver pair. Nothing else changes, so the difference between the two Sionna modes is vegetation alone.

All three modes share the towers, grid, users and handoff logic. Only the propagation changes.

Vegetation is P.833 loss added after the ray trace

- **SionnaRT has no vegetation inside.** Sionna 2.2.0 has no vegetation material, foliage flag or enable switch, so trees cannot simply be turned on.
- **Foliage depth per pair:** for each tower–receiver straight line, the depth through the crowns of the 1,919 surveyed DataSF street trees – crown discs of radius 2–5 m, height 2–12 m from trunk diameter.
- **P.833 loss:** $A = Am \cdot (1 - \exp(-d \cdot \gamma / Am))$, with $(\gamma, Am) = (0.55, 38)$ interpolated at 3.6 GHz – assumed, uncalibrated. Applied to LOS pairs only, so it is a **lower bound**: foliage on reflected paths is not counted.
- **Cross-check:** a trial with crowns as solid Sionna geometry gave a larger effect – an upper bound. The truth lies between the two.



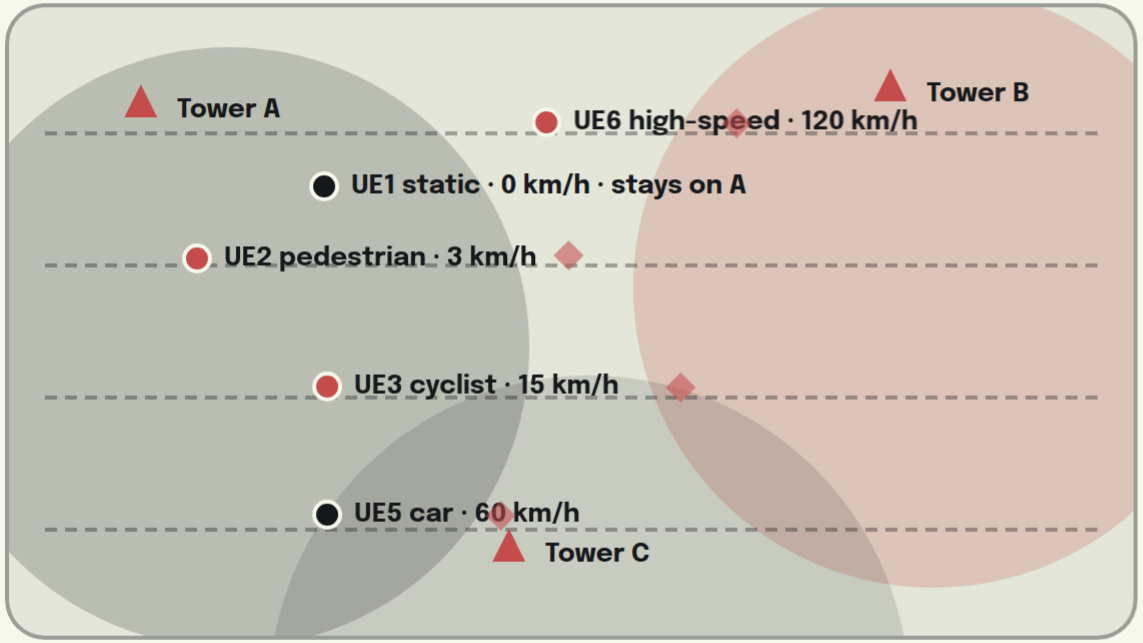
Handoffs are triggered by the UE's CQI report

UE1 – static	**0 km/h**
UE2 – pedestrian	**3 km/h**
UE3 – cyclist	**15 km/h**
UE4 – car	**30 km/h**
UE5 – car	**60 km/h**
UE6 – high-speed	**120 km/h**

Each user has a position and a velocity, and is scored at sector level against all nine sectors on every render.

- **SINR, computed per sector:** each sector's RSRP is set against the RSRP of all the other sectors plus thermal noise, so the interference from every neighbour counts. The map is coloured by best server, and each user's table row shows serving sector, SINR and CQI.
- **CQI, reported by the UE:** the UE maps its SINR to a CQI from 0 to 15 – SINR quantised in roughly 2 dB steps – and reports it, as a UE does in a real network.
- **Handoff trigger:** a handover fires when a neighbour's CQI beats the serving cell's CQI. The handoff region is where the top two CQIs are within 1, with a floor of CQI ≥ 1 – CQI 0 is out of range, no usable link.
- **Doppler and ICI, calculated:** Doppler = speed × frequency / c per user; ICI ratio = $(\pi \cdot f_D / \Delta f)^2 / 3$, folded into effective SINR and the modelled capacity.

Play movement: users cross the map and hand over live

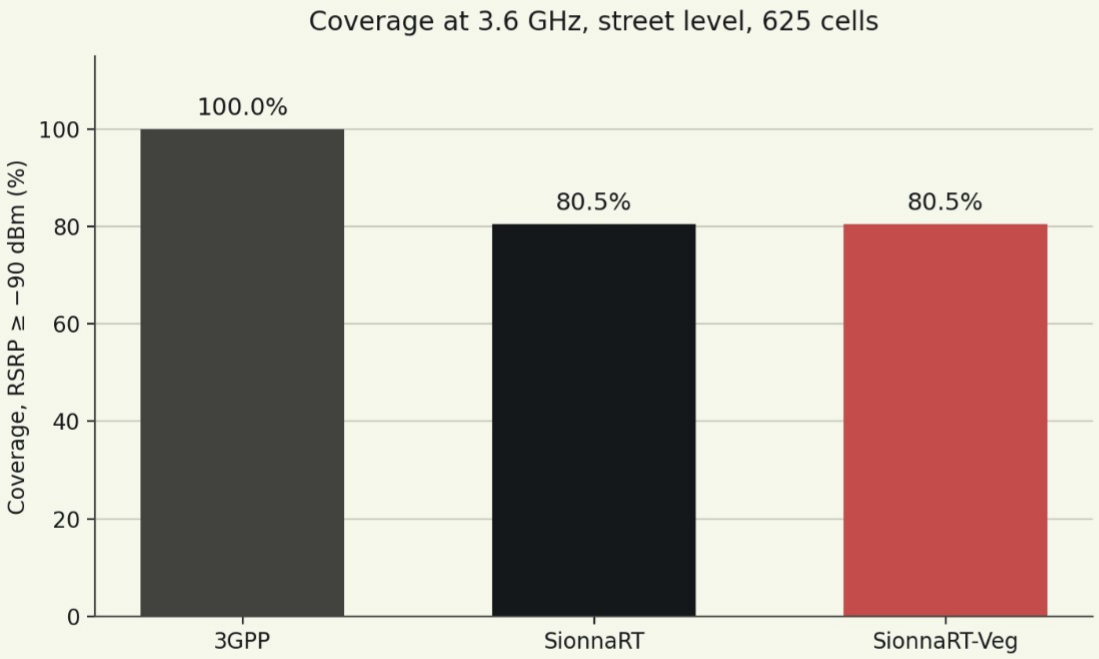


Serving: A → Handoff → Serving: B → Handoff → Serving: C

Pulsing diamonds are handoff points on each route, where best and second-best RSRP come within that user's threshold.

- **Press Play movement in the twin:** the button becomes Pause movement, and each user advances by velocity × 0.5 per frame, bouncing at the tile edges (0 and 1,000 m).
- **The map redraws as they move:** colour means best server, dots travel their lanes, and the serving sector flips when another sector's RSRP passes the current one.
- **Handoff events, marked live:** white dots mark the inter-tower region, yellow squares the intra-tower sector region – wherever best minus second-best is inside that user's threshold: 3 dB static and pedestrian, 4 dB cyclist, 5 dB at 30 km/h, 6 dB at 60 km/h, 8 dB at 120 km/h. Inside it a handover fires; ping-pong is possible.
- **The table follows the movement:** serving sector, RSRP, second sector and difference, handoff flag, Doppler, ICI and modelled capacity.

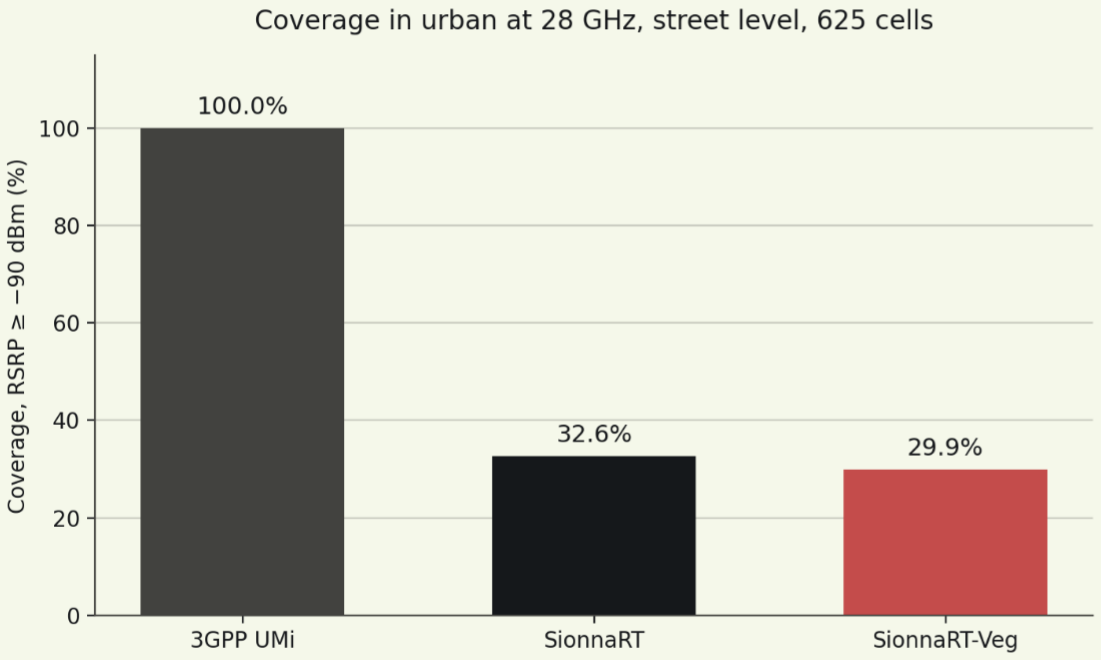
Trees are a tail effect at 3.6 GHz, not a coverage effect



- **Median best-server RSRP:** −31.9 dBm (3GPP), −72.7 dBm (SionnaRT), −72.7 dBm with trees. The ray-traced map is far weaker because it sees the real buildings.
- **Handoff region (CQI trigger, CQI ≥ 1 floor):** 12.6% of cells on the 3GPP map, 9.5% ray-traced, 9.7% with trees.
- **Tree loss per cell:** median 0.0 dB, mean −0.18 dB, 4.0% of cells lose more than 1 dB, worst cell −11.7 dB. One user, the cyclist, drops 5.2 dB; the other five are unchanged.

All figures are ray-traced, modelled or simulated – none are measured.

At 28 GHz, small cells in urban cover a third of the tile



- **The layout:** 16 small cells on a 4×4 grid, 250 m apart, 8 m lamppost height, 3 sectors per cell, tilt 4°, EIRP 40 dBm – same downtown San Francisco tile, street level.
- **Median best-server RSRP:** −50.9 dBm (3GPP UMi), −97.4 dBm (SionnaRT), −98.1 dBm with trees. 16.6% of cells have no ray-traced path, and only 19.4% have line of sight to any small cell.
- **Trees matter more here:** median change 0.0 dB, mean −1.35 dB, 7.2% of cells lose more than 1 dB, worst cell −41.4 dB – coverage falls 2.7 points, to 29.9%. LiDAR canopy sensitivity: 26.9%.
- **Densification helps only modestly:** 16 cells lift coverage from 25.3% (one 30 m tower at 28 GHz) to 32.6% – at 8 m the cells are blocked most of the time.

All figures ray-traced, modelled or simulated – none measured. P.833 (γ, Am) = (2.20, 50), interpolated, uncalibrated.

The same city, three honest answers

3GPP gives the planning sketch, SionnaRT gives the street as the buildings make it, and SionnaRT-Veg adds the trees as a stated lower bound. The limits are stated too: SionnaRT data exist at 3.6 GHz only, the grid is 40 m, and P.833 is interpolated and LOS-only.

https://ysrao.github.io/twin5GNR-multicell-3gpp/twin5GNR-multicell-SionnaRT-Veg.html